

Decoupling-Oriented Coordination of Paralleled VSGs With Multi-Agent Reinforcement Learning

Yuang Wei, Yixiao Wang, and Zhuang Xu

Department of Electrical and Electronic Engineering
University of Nottingham Ningbo China, Ningbo, China
John.XU@nottingham.edu.cn

Abstract—Coordination among paralleled virtual synchronous generators (VSGs) cannot be inferred from average-frequency recovery alone because aggregate support and relative motion are different physical coordinates. This paper evaluates a direct inertia/damping multi-agent reinforcement learning (MARL) formulation under a guard-first contract that separates common-frequency restoration, off-diagonal common-differential response, disturbance-driven differential energy, and command magnitude/variation. Each of four actors writes only its own bounded, slew-limited inertia/damping pair. A sequence of matched repairs makes the learning trajectories valid and removes selected guard failures, but none of the reported valid direct- M/D configurations satisfies the complete physical contract. We then estimate a dataset-level algorithmic source contrast with a prospectively frozen eight-arm factorial, six training seeds, and a same-time row-permuted placebo. Authentic neighbour sources yield a signed geometric effect of -2.45% for the actor factor (unfavourable to the authentic source) and $+4.52\%$ for the critic factor; exact sign-flip tests with Holm control did not establish either effect above the 10% materiality bar. This is a bounded “not established above 10%” result, not evidence of zero communication value. A distinct structured energy-port controller passes its own registered development and unseen-bank contract. This related, non-pooled witness keeps the negative MARL conclusion specific to the tested action, learner, and 43,200-step schedule.

Keywords—virtual synchronous generator, multi-agent reinforcement learning, frequency coordination, virtual inertia, factorial evaluation

I. INTRODUCTION

Virtual synchronous generators reproduce swing-like behaviour in converter controls and expose configurable inertia and damping [1], [2]. When several VSGs operate in parallel, however, their average frequency and their relative motion are not the same control objective. Consensus and mutual-damping controllers address frequency restoration and oscillatory interaction with explicit information structures [3]–[7]. The distinction is visible in a localized disturbance: a fleet-wide inertia change directly addresses aggregate support but not inter-unit motion, whereas spatially different commands target relative motion and may create additional command variation. Figure 1 makes these two readouts explicit.

Learning-based VSG controllers adapt virtual parameters without requiring an explicit plant model [8]–[15]. Here, we ask a different question from success-oriented controller design: whether neighbour-source access adds differential coordination after actor, critic, reward, execution semantics, and

physical endpoints are matched. Three evaluation problems remain when such controllers are interpreted as coordination mechanisms. First, a scalar reward or an average-frequency metric can mix common support with differential motion. Second, a favourable endpoint can be obtained on commands that violate amplitude, slew, or command-variation requirements. Third, a nominal message/no-message comparison can change actor, critic, reward, or source-distribution access at the same time. Seed variation adds further uncertainty to small-run reinforcement-learning studies [16]. These issues do not imply that MARL is unsuitable; they mean that its incremental coordination value must be defined before it is measured.

Our goal is therefore to evaluate whether distributed direct- M/D MARL adds physically valid differential coordination beyond fixed common support and a strong deterministic comparator. We define decoupling through two finite-window physical endpoints: off-diagonal response between arithmetic common and differential coordinates, and differential-frequency energy across signed common, differential, and localized probes. Endpoint improvement is interpreted only after common-frequency, peak, RoCoF, action, slew, bound, and execution-validity guards pass. The paper thus studies a new evaluation setting rather than presuming a successful learned controller.

This setting creates three technical challenges. The physical coordinates and guards must remain consistent across deterministic and learned controllers. Actor commands must be compared after their executed-action semantics are made valid, because raw neural outputs and slew-projected plant inputs are different objects. Finally, neighbour information must be tested against a placebo that preserves time and marginal structure; otherwise a donor distribution can be mistaken for semantic message value.

We address these challenges with a guard-first evaluation pipeline. A modified Kundur system in ANDES [17] provides signed common/differential probes and localized disturbances. Direct- M/D learning variants share bounded per-VSG actions, fixed budgets, final-checkpoint evaluation, and a strong local neighbour comparator. Representative interventions then test action projection, value scaling, and action penalties without concealing failed runs. The final source experiment crosses actor source, critic source, and reward access, using authentic same-time neighbour rows or a prospectively frozen row permutation. A separate energy-port controller is retained

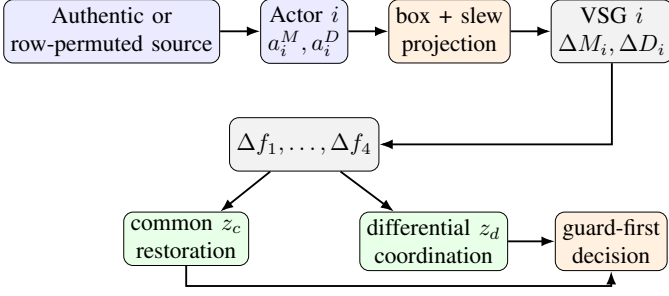


Fig. 1. The evaluation contract separates the information source, executed direct- M/D action, common-frequency support, and differential coordination. Endpoint improvement is accepted only after the physical and command guards pass.

only as a non-pooled witness for its distinct action object and endpoint contract.

This work makes three contributions. First, we formulate a physical coordination contract that separates common support, differential response, and empirical command guards (Section II). Second, we provide a matched evaluation showing that successive direct- M/D learner repairs move, but do not close, the residual guard gap (Sections III–IV). Third, a six-seed source factorial does not establish an actor or critic effect above a prospectively frozen 10% materiality bar, while a distinct structured controller bounds the negative conclusion to the tested formulation (Sections IV–V).

II. PHYSICAL COORDINATION CONTRACT

A. Control object and coordinates

The model contains four swing-equation VSG proxies. Their rotor-angle and frequency states satisfy

$$\dot{\delta}_i = 2\pi f_n(\omega_i - 1), \quad M_i \dot{\omega}_i = \tau_{m,i} - \tau_{e,i} - D_i(\omega_i - 1), \quad (1)$$

where $f_n = 60$ Hz, $M_i = 2H_i$ is the ANDES startup-time parameter in seconds, and D_i is its per-unit damping coefficient. Actor i independently produces $a_i = [a_i^M, a_i^D]^\top \in [-1, 1]^2$ every 0.2 s. Each component is limited to a change of 0.25 per update. Equation (2) decodes the executed action into local virtual inertia and damping:

$$\begin{aligned} \Delta q(a) &= \begin{cases} 600a, & a \geq 0, \\ 200a, & a < 0, \end{cases} \\ M_i[k] &= \max\{20, M_i^0 + \Delta q(a_i^M[k])\}, \\ D_i[k] &= \max\{10, D_i^0 + \Delta q(a_i^D[k])\}. \end{aligned} \quad (2)$$

The learner bank spans $M_i^0 \in [140, 260]$ s and $D_i^0 \in [50, 150]$ on the same per-unit model base. No actor directly injects active power.

Let $\Delta f \in \mathbb{R}^4$ denote the physical 60 Hz frequency deviations. We use the arithmetic common coordinate $z_c = 1^\top \Delta f / 4$. Equation (3) defines the differential coordinate:

$$z_d = T_d \Delta f, \quad T_d = \begin{bmatrix} 1/2 & 1/2 & -1/2 & -1/2 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}. \quad (3)$$

These are registered arithmetic coordinates, not identified eigenmodes. For signed probe kind $k \in \{c, d, \ell\}$, define the odd response $\tilde{f}_k[n] = (\Delta f_k^+[n] - \Delta f_k^-[n])/2$, $z_{c,k}[n] = 1^\top \tilde{f}_k[n]/4$, and $z_{d,k}[n] = T_d \tilde{f}_k[n]$. With signed-pair magnitude ε_k , the two lower-is-better endpoints are

$$J_{\text{cross}} = \frac{\Delta t}{\varepsilon_c^2} \sum_{n=1}^N \frac{\|z_{d,c}[n]\|_2^2}{3} + \frac{\Delta t}{\varepsilon_d^2} \sum_{n=1}^N z_{c,d}^2[n], \quad (4)$$

$$J_d = \sum_{k \in \{c, d, \ell\}} \frac{\Delta t}{\varepsilon_k^2} \sum_{n=1}^N \frac{\|z_{d,k}[n]\|_2^2}{3}, \quad (5)$$

where $N = 30$ and $\Delta t = 0.2$ s. The positive input vectors are $u_c = \varepsilon_c[1, 1, 1, 1]^\top/4$, $u_d = \varepsilon_d[1, 1, -1, -1]^\top/4$, and $u_\ell = \varepsilon_\ell e_j$; each has a matched negative pair. The frozen profile table supplies every ε_k and localized-load index. Both endpoints have units $\text{Hz}^2 \text{s/pu}^2$ and are discrete L_2 response energies, not joules, storage functions, or Lyapunov functions. Thus J_d includes the common, differential, and localized pairs. The arithmetic z_c is not an inertia-weighted centre of inertia, and T_d is not a modal basis.

B. Guard-first decision

A controller is interpreted only after its trajectories complete and its identity, bounds, slew, saturation, and independent per-VSG action checks pass. Common-frequency integral, worst-unit peak, and RoCoF must remain within 103% of the deterministic reference, while executed-action RMS and total variation must remain within 110%. These empirical guards apply to the tested bank; they are not stability or safety certificates. The margins were frozen before confirmatory analysis and encode a tighter non-degradation tolerance for grid-response quantities than for command summaries. Training reward is never used to override a failed physical guard. Safe and constrained reinforcement learning similarly distinguishes an optimization penalty from verified closed-loop constraints [18], [19].

RoCoF is the worst absolute one-step finite difference of physical VSG frequency, including the initial-to-first-sample interval, divided by Δt . Command RMS is taken over all eight executed normalized components; total variation sums the per-step mean absolute component change from zero within each signed record.

Let $v = [\Delta M_1, \Delta D_1, \dots, \Delta M_4, \Delta D_4]^\top$ denote the decoded direct- M/D command vector. At the synchronous power-balanced equilibrium, a folded index-1 DAE linearization gives the channel in (6):

$$B_{v,r} = f_v - f_y g_y^{-1} g_v = 0 \quad (6)$$

for all eight direct- M/D command columns. At this equilibrium, the torque residual and $\omega_i - 1$ vanish in (1), while the network algebraic equations contain no direct M/D column; hence $f_v = g_v = 0$ and the folded channel is zero. This is an equilibrium first-order authority result: it does not remove the off-equilibrium multiplicative effect of M and D , and it is not treated as the sole cause of learning failure.

III. MARL AND CONFIRMATORY PROTOCOL

A. Plant, comparator, and learners

All direct- M/D trajectories use ANDES 2.0.0 on a modified Kundur two-area system at 60 Hz, with 0.2 s controller updates over a 6 s post-disturbance window. Deterministic comparator selection uses two development and four fixed evaluation profiles; learner training and confirmatory evaluation use a separate bank of four development and four fixed evaluation profiles. Both banks vary loading, VSG parameter mismatch, disturbance location, sign, magnitude, and spatial distribution before execution. A bank of nine local-neighbour dynamic M/D laws is selected on development data. The selected law reduces off-diagonal and disturbance differential energy by 60.79% and 64.13%, respectively, relative to zero action while passing every development guard. It is therefore a stronger comparator than zero action.

The learning studies include parameter-shared multi-agent TD3 and adapted SAC families. TD3 uses delayed policy updates and twin critics [20]; the multi-agent critic follows centralized-training, decentralized-execution practice [21]. Each actor receives its local active-power, frequency, and RoCoF features plus two ring-neighbour frequency/RoCoF pairs. No-message rows retain the same dimension but mask the four neighbour entries in every actor path. Training uses 43,200 physical interaction steps per seed and deterministic final checkpoints. Executed actions, rather than raw network outputs, are used by the physical evaluation. Neighbour rows are sampled synchronously at each decision instant; delay, packet loss, and quantization are not modelled. The critic-source factor affects centralized training only; evaluation deploys the deterministic actor with its frozen actor-source mapping and no critic access.

The confirmatory factorial freezes the projected adapted-SAC learner. Its Gaussian-tanh actor and twin-Q critic each use four 128-unit hidden layers, learning rate 3×10^{-4} , $\gamma = 0.99$, $\tau = 0.005$, a 10,000-sample replay buffer, and batch size 256. Automatic entropy temperature is bounded to $[0.005, 5]$; one update follows each environment step after buffer warm-up, and evaluation uses the deterministic actor mean. The reward is the unchanged weighted sum $r_i = 100r_{f,i} + 50r_{abs,i} + 0.0056r_M + 0.0056r_D$. Here $r_{f,i}$ penalizes squared disagreement among the local frequency and two ring neighbours, $r_{abs,i}$ penalizes squared local frequency deviation after conversion to physical hertz, and r_M, r_D penalize the squared fleet-mean decoded pre-clamp command increments normalized by 600. The reward-access factor switches only the neighbour terms in $r_{f,i}$.

B. Representative repair ladder

The archived development line includes invalid, negative, and partial repairs; Table I reports representative valid stages rather than an all-attempt flow diagram. Target normalization suppresses critic-scale divergence and removes command-summary failures, but common-frequency and worst-peak guards remain. Execution-layer projection makes adapted SAC

TABLE I
REPRESENTATIVE VALID LEARNER REPAIRS ON THE DIRECT- M/D OBJECT

Intervention	Common/peak guards	Action guards	Full gate
Value-target normalization	fail	pass	fail
Projected adapted SAC	pass (message)	fail	fail
SAC + RMS penalty	fail	RMS only	fail

trajectories valid and improves both endpoints, but executed-action RMS and variation remain above the comparator. Penalizing action RMS closes that guard while frequency restoration and action variation regress. No stage passes the full direct- M/D contract.

C. Clean neighbour-source factorial

The confirmatory experiment crosses actor source $\{N, P\}$, critic source $\{N, P\}$, and reward access $\{0, 1\}$, producing eight arms for seeds 401–406. N is the authentic same-time ring-neighbour source. P retains each recipient's local columns but replaces neighbour columns with the authentic row of the next actor, $\rho(i) = (i + 1) \bmod 4$. This placebo preserves time and uses no exogenous donor bank. More precisely, it leaves the three recipient-local columns unchanged and row-permutes only the four neighbour columns. The permutation is fixed-point-free: for every neighbour column, its four-value multiset and the multiset of ordered neighbour tuples are identical under N and P , while no P tuple is the authentic neighbour tuple of its recipient. Reward access independently switches the neighbour terms in the coordination reward; the reported source effects marginalize over this factor.

All 48 training cells completed 43,200 steps. Sixteen cells were hash-identical carryovers accepted only after per-cell identity and completeness checks; the other 32 were trained afresh. Each carryover had to match the arm, seed, factor assignment, complete 43,200-step manifest, reward hash, and base-state hash. Half and final checkpoints produced 16 complete evaluation jobs, and all 48 final cells were valid. The manifest-bound reproducibility supplement provides the frozen input banks, all stage-arm-seed-profile endpoints, interaction-sensitive contrasts, carryover provenance, curve-stability rows, topology variants, and the complete confirmatory run flow.

Let L be disturbance differential energy on one evaluation profile. For each seed, the actor-source effect is the equal-weight mean of $\log[L(P, c, r)/L(N, c, r)]$ over matched critic sources, reward-access values, and profiles; the critic-source effect is defined analogously over matched actor sources. Positive values therefore favour the authentic source. The primary estimand is the across-seed mean at the final checkpoint. The materiality threshold is $\log(1.10)$. At this boundary, we assume joint sign invariance of the six paired seed effects and monotone symmetric location shifts. We then enumerate all 2^6 seed-level sign flips, test each factor one-sided, and apply Holm control across the two tests [22]. The descriptive 95%

intervals are percentile intervals from all 6^6 ordered seed-level resamples. As an asymmetric-distribution sensitivity, exact binomial sign tests for a seed-median effect above the same bar count 1/6 actor and 2/6 critic effects above it ($p = 0.9844$ and 0.8906); this is a different estimand from the primary mean. Half/final direction consistency and leave-one-seed-out means are also reported. The frozen curve-stability check compares median absolute actor/critic losses in the final two training deciles and requires an absolute log-ratio no larger than $\log(1.25)$; it does not establish that the source-effect estimand has plateaued.

IV. RESULTS

Table II collects selected valid evidence stages. Invalid trajectories and incomplete checkpoints are not numerically pooled with valid rows. The table also keeps the direct- M/D and energy-port objects separate: their ratios have different actuators, estimators, banks, and references and are therefore not pooled.

A. Direct- M/D learning remains guard limited

The deterministic development-selected law is also selected by a finite outcome-seeing oracle on all four evaluation profiles, so the original nine-law family provides no additional profile-conditional headroom. A later 350-schedule time-varying family contains one development-selected schedule that is guard-clean on two of four fixed evaluation profiles. Thus direct M/D authority is not empty, but the finite evidence does not supply a uniform transferable target for the learner.

Among the valid direct- M/D learner stages, projected adapted SAC provides the largest guard-valid endpoint movement. Its message arm reaches 0.635 and 0.590 times the deterministic reference on the disturbance and off-diagonal endpoints and passes the common-frequency and worst-peak guards in all 20 seed-profile blocks. RoCoF still fails in 4/20 blocks, and both command-summary guards fail in all 20 blocks. Adding an RMS penalty eliminates the RMS failures but moves the endpoint ratios to 1.229 and 1.221 and produces 20/20 common-frequency and 18/20 worst-peak failures, together with 7/20 RoCoF failures. The interventions therefore exchange guard failures rather than producing a guard-clean direct- M/D learner.

Figure 2 gives a non-data-selected time-domain view of this exchange on the first registered evaluation profile under its common-positive disturbance. Projection reduces the common-frequency deviation but requires substantially larger executed actions. The RMS penalty returns action magnitude near the deterministic trace while the frequency deviation becomes larger than the comparator. This trace illustrates the mechanism; the complete gate counts in Table II carry the inference.

B. No source effect above the materiality bar is established

Figure 3 reports the confirmatory source effects. The actor factor has final marginal mean log effect -0.0248 , corresponding to a -2.45% geometric effect unfavourable to

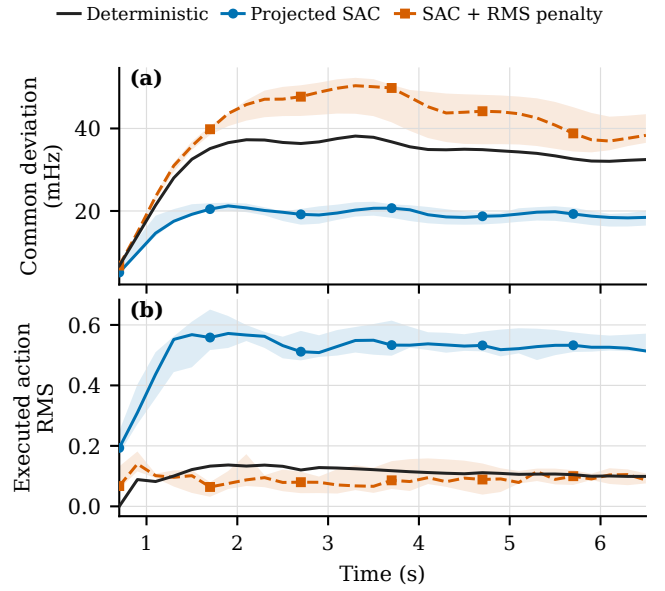


Fig. 2. Projection improves common-frequency response but raises command magnitude; the RMS penalty reverses the trade-off. Panel (a) shows 1000(60– f) and panel (b) shows the RMS of the eight executed normalized action components. Learned curves are medians across seeds 401–405; shading spans their range, while the deterministic comparator is a single trace. The profile and disturbance are fixed independently of the plotted outcome.

the authentic source, with descriptive 95% log-scale interval $[-0.0882, 0.0450]$. Only one of six seed effects is positive. The critic factor has mean log effect 0.0442, or $+4.52\%$, with log-scale interval $[-0.0092, 0.0908]$ and five positive seed effects. The exact materiality p values are 0.984375 and 0.953125; neither Holm test rejects. No marginal half/final direction flip is recorded, and all loss rows pass the stated curve-stability rule. However, actor and critic marginal effects contract from -13.23% and $+23.59\%$ at half budget to -2.45% and $+4.52\%$ at final budget. Leave-one-seed-out mean log effects remain negative for the actor (-0.0556 to -0.0016) and positive for the critic (0.0284 to 0.0671).

The marginal directions are not component-wise effects. Conditioning on critic source changes the final actor contrast from $+14.21\%$ for authentic critic rows to -16.68% for placebo critic rows. Conditioning on reward access changes the critic contrast from -7.36% to $+17.91\%$. These unregistered interactions are descriptive and rule out interpreting the marginals as stable isolated actor or critic benefits.

The supported wording is narrow: within this six-seed learner, bank, and projector, authentic same-time neighbour sources versus the row-permuted placebo do not establish an effect above 10% in either factor. The contrast is a dataset-level algorithmic source contrast: the permutation also changes recipient-conditioned coupling and source roles, so pure semantic information value is not identified. The six-seed design is low resolution, and smaller effects are not excluded.

Sixteen hash-identical carryovers all occupy actor-source- N cells; no same-round retraining comparison was performed.

TABLE II
VALID EVIDENCE LADDER AND THE CLAIM RETAINED FROM EACH STAGE

Stage	Object and evaluation unit	Valid result	Retained claim
Deterministic finite bank	Nine direct- M/D laws; development selection and four fixed evaluation profiles	Selected law lowers the two development endpoints by 60.79% and 64.13% versus zero; the outcome-seeing nine-law oracle selects the same law on all four profiles	Strong comparator, but no extra profile-conditional headroom inside this finite family
Value-normalized TD3	Direct- M/D learner; matched seed-profile bank	Critic scale and command-summary guards recover, while common-frequency and worst-peak guards remain failed	Numerical stabilization alone does not close the physical contract
Projected adapted SAC	Direct- M/D learner; 20 message-arm seed-profile blocks	All trajectories are valid; endpoint ratios are 0.635/0.590, common-frequency and worst-peak guards pass, RoCoF fails in 4/20 blocks, and both command-summary guards fail in every block	Partial endpoint movement, not a guard-clean controller
SAC with RMS penalty	Same direct- M/D object and bank	RMS failures fall to 0/20, but action variation remains failed; endpoint ratios regress to 1.229/1.221, with common-frequency, worst-peak, and RoCoF failures in 20/20, 18/20, and 7/20 blocks	A measured trade-off, not a repair
Clean source factorial	Eight direct- M/D arms, six seeds, matched profiles	Actor and critic geometric effects are -2.45% and $+4.52\%$; neither clears the 10% materiality test with Holm control	Neither source factor clears the 10% materiality test; zero effect is not established
Structured energy port	Feasible power-residual action; frozen unseen and topology-variant banks	Deterministic controller passes the endpoints on the unseen bank and all ten equilibrium-sound topology variants; residual SAC adds no endpoint or message increment	The target is nonempty for a distinct action object; no numerical ranking against direct M/D

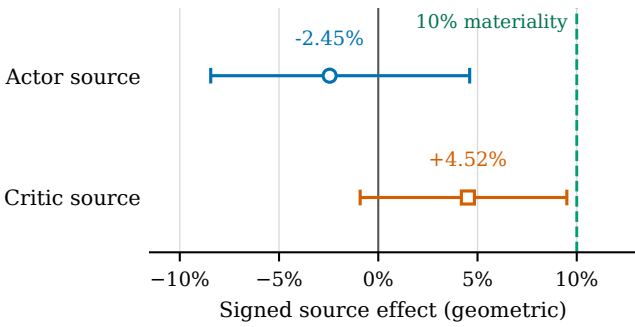


Fig. 3. Neither authentic neighbour-source factor establishes an effect above the 10% materiality bar. Points show geometric effects and bars transform the descriptive bootstrap intervals from the paired log scale; inference uses the exact six-seed sign-flip tests with Holm control.

The reuse audit establishes file/configuration identity and arithmetic replay, but cannot empirically exclude a batch-by-actor-source effect. Therefore the signed directions remain descriptive for the sealed dataset rather than independent repeatability claims.

C. A distinct structured controller bounds the negative result

The energy-port companion replaces direct parameter modulation with a state-dependent feasible power-headroom map and a frozen 0.4 Hz ring-edge bandpass. Because its actuator, estimator, bank, and reference differ, its ratios are not pooled with the MARL results. After development selection, $K = 3.5$ is frozen and attains $r_d = 0.938218$ and $r_{\text{cross}} = 0.793730$ on a one-use unseen bank, with every guard passing. It also passes all ten equilibrium-sound members of a frozen topology-variant bank: the nominal network, four inter-area line outages, and five line-reactance scalings. Two VSG-tie outages fail the pre-performance initialization/equilibrium-information gate and are excluded from performance scoring. Across three additional condition/parameter blocks, two pass and a relaxed-

inertia/heavy-damping block misses the differential ceiling with $r_d = 0.9712$ while retaining all guards.

A baseline-anchored residual SAC then preserves the band-pass passes on the ten sound variants but never improves either endpoint and shows no message increment. This companion supports two bounded statements: a different energy-port action object attains its own matched endpoint contract on finite banks, and adding a learner to that strong structure provides no measured increment in the tested configuration.

V. INTERPRETATION AND LIMITATIONS

The combined evidence localizes, but does not uniquely identify, the negative result. On the gauge-fixed, fixed-mode direct- M/D model, the Schur-folded Jacobian $B_{v,r} = f_v - f_y g_y^{-1} g_v$ is zero for all eight direct- M/D columns at the synchronous power-balanced equilibrium, so no additive first-order reduced-state channel is present there. Converged local physical-parameter derivatives establish mixed state-parameter dependence away from equilibrium; the normalized policy and asymmetric decoder are only piecewise differentiable at zero, so the local interpretation is one-sided. The sealed 30-step finite-amplitude ladder is quadratic-leading in all 12 tested blocks but neither separates mixed from pure quadratic terms nor establishes a global Taylor law. These facts are consistent with, but do not causally identify, a difficult residual-learning interface. The repair ladder also shows that critic scale, command execution, command summaries, and frequency restoration are distinct failure axes: correcting one does not imply that the others close. The energy-port result suggests that the choice of physical action basis can matter more than increasing learner freedom, but the unmatched objects prevent a causal ranking.

The conclusions cover one phasor-domain simulator, one modified Kundur plant, registered finite banks, the tested learner families, the fixed 43,200-step schedule, and at most six training seeds. Arithmetic common/differential coordinates are neither modal eigenvectors nor inertia-weighted COI coordinates; time-domain guards are not stability or safety

certificates. The topology variants do not support arbitrary graph or size generalization, and no EMT, HIL, hardware, or field claim is made. Most importantly, failure to establish a 10% source effect is not an equivalence test and does not imply zero communication value. Communication is idealized as synchronous and noiseless; delay, loss, and quantization remain outside the tested contract.

VI. CONCLUSION

This paper has evaluated decoupling-oriented MARL coordination under a guard-first physical contract. With matched action semantics and a clean row-permuted placebo, the reported direct- M/D configurations have neither closed the full coordination contract nor established an actor or critic marginal source effect above 10% within six seeds and 43,200 steps. A separate energy-port action object attains its own matched contract on bounded banks; it does not establish direct- M/D attainability. Future work should independently repeat the source factorial, test physically structured residual actions, and require held-out networks before topology-general claims.

REFERENCES

- [1] Q.-C. Zhong and G. Weiss, "Synchronverters: Inverters that mimic synchronous generators," *IEEE Transactions on Industrial Electronics*, vol. 58, no. 4, pp. 1259–1267, 2011.
- [2] S. D'Arco and J. A. Suul, "Virtual synchronous machines—classification of implementations and analysis of equivalence to droop controllers for microgrids," in *Proc. IEEE Grenoble PowerTech*, 2013, pp. 1–7.
- [3] Y. Liu, Y. Wang, H. Xu, H. Zhou, H. Liu, and Y. Peng, "Modeling and parameter analysis for consensus based secondary control of parallel virtual synchronous generators," *Energy Reports*, vol. 6, pp. 1462–1475, 2020.
- [4] M. Shi, X. Chen, J. Zhou, Y. Chen, J. Wen, and H. He, "Frequency restoration and oscillation damping of distributed VSGs in microgrid with low bandwidth communication," *IEEE Transactions on Smart Grid*, vol. 12, no. 2, pp. 1011–1021, 2021.
- [5] S. Fu, Y. Sun, L. Li, Z. Liu, H. Han, and M. Su, "Power oscillation suppression of multi-VSG grid via decentralized mutual damping control," *IEEE Transactions on Industrial Electronics*, vol. 69, no. 10, pp. 10 202–10 214, 2022.
- [6] X. Gao, D. Zhou, A. Anvari-Moghaddam, and F. Blaabjerg, "An adaptive control strategy with a mutual damping term for paralleled virtual synchronous generators system," *Sustainable Energy, Grids and Networks*, vol. 38, p. 101308, 2024.
- [7] X. Wang, Z. Zhong, W. Jiang, and X. Zhao, "Consensus-based secondary frequency control for parallel virtual synchronous generators," *Sustainable Energy, Grids and Networks*, vol. 38, p. 101341, 2024.
- [8] Q. Yang, L. Yan, X. Chen, Y. Chen, and J. Wen, "A distributed dynamic inertia-droop control strategy based on multi-agent deep reinforcement learning for multiple paralleled VSGs," *IEEE Transactions on Power Systems*, vol. 38, no. 6, pp. 5598–5612, 2023.
- [9] C. Lu and X. Zhuan, "Adaptive control for virtual synchronous generator parameters based on soft actor critic," *Sensors*, vol. 24, no. 7, p. 2035, 2024.
- [10] Z. Benhmouch, S. Moufid, A. Ait-Omar, A. Abbou, H. Laabassi, M. Kang, C. Chatri, I. H. O. Ali, H. Bouzekri, and J. Baek, "A novel reinforcement learning policy optimization based adaptive VSG control technique for improved frequency stabilization in AC microgrids," *Electric Power Systems Research*, vol. 230, p. 110269, 2024.
- [11] D. Zhang, J. Zhang, Y. He, T. Shen, and X. Liu, "Adaptive control of VSG inertia damping based on MADDPG," *Energies*, vol. 17, no. 24, p. 6421, 2024.
- [12] S. Kang, Y. Jung, D. You, and G. Jang, "Enhancing frequency stability with decentralized adaptive control using multi-agent deep reinforcement learning of multi-VSGs," *International Journal of Electrical Power & Energy Systems*, vol. 172, p. 111374, 2025.
- [13] J. Li, C. Liu, H. Liu, H. Liu, L. Shao, and L. Zhu, "Adaptive VSG control method based on reinforcement learning strategy optimization," *Electric Power Systems Research*, vol. 257, p. 112997, 2026.
- [14] M. Ikram, A. Aziz, and D. Habibi, "A novel multi-agent deep reinforcement learning framework for fast frequency response in inverter-based hybrid power plants," *Sustainable Energy, Grids and Networks*, vol. 46, p. 102282, 2026.
- [15] X. Ding and J. Cao, "Deep and reinforcement learning in virtual synchronous generator: A comprehensive review," *Energies*, vol. 17, no. 11, p. 2620, 2024.
- [16] R. Agarwal, M. Schwarzer, P. S. Castro, A. C. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 29 304–29 320.
- [17] H. Cui, F. Li, and K. Tomovic, "Hybrid symbolic-numeric framework for power system modeling and analysis," *IEEE Transactions on Power Systems*, vol. 36, no. 2, pp. 1373–1384, 2021.
- [18] D. Tabas and B. Zhang, "Computationally efficient safe reinforcement learning for power systems," in *Proc. American Control Conference*, 2022, pp. 3303–3310.
- [19] H. Shuai, B. She, J. Wang, and F. Li, "Safe reinforcement learning for grid-forming inverter based frequency regulation with stability guarantee," *Journal of Modern Power Systems and Clean Energy*, vol. 13, no. 1, pp. 79–86, 2025.
- [20] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in *Proc. 35th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 80, 2018, pp. 1587–1596.
- [21] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 6379–6390.
- [22] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.